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# B.TECH/ M.Tech (Integrated) DEGREE EXAMINATION, MAY 2025

Sixth Semester

21CSS303T - DATA SCIENCE

(For the candidates admitted during the academic year 2021-2022 to 2024-2025)

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## Note:

- Part - A should be answered in OMR sheet within first 40 minutes and OMR sheet should be handed over to hall invigilator at the end of 40<sup>th</sup> minute.
- Part - B and Part - C should be answered in answer booklet.

Time: 3 hours

Max. Marks: 75

## PART - A (20 × 1 = 20 Marks)

Answer all Questions

Marks BL CO

- |                                                                                                                                                                 |   |   |   |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---|
| 1. Which of the following is used to create an identity matrix in NumPy?<br>(A) numpy.identity() (B) numpy.eye()<br>(C) numpy.ones() (D) numpy.zeros()          | 1 | 1 | 1 |
| 2. Which of the following is not a facet of data science?<br>(A) Data Engineering (B) Data Visualization<br>(C) Machine Learning (D) Operating System           | 1 | 1 | 1 |
| 3. In Pandas, what is the output of df.describe()?<br>(A) Correlation matrix (B) Summary statistics<br>(C) Shape of the DataFrame (D) Shape of the Data_Frame   | 1 | 1 | 1 |
| 4. Which Pandas object is best suited for labeled one-dimensional data?<br>(A) DataFrame (B) Panel<br>(C) Series (D) Array                                      | 1 | 1 | 2 |
| 5. What does the .drop() function do in Pandas?<br>(A) Drops the database (B) Drops entries/columns from DataFrame<br>(C) Drops Pandas library (D) Drops arrays | 1 | 1 | 3 |
| 6. Which technique is NOT used for data acquisition?<br>(A) Web scraping (B) SQL queries<br>(C) Image rendering (D) APIs                                        | 1 | 1 | 3 |
| 7. In NumPy, which method is used to join two arrays?<br>(A) numpy.add() (B) numpy.concatenate()<br>(C) numpy.merge() (D) numpy.connect()                       | 1 | 1 | 1 |
| 8. What is data wrangling primarily used for?<br>(A) Enhancing images (B) Cleaning and transforming data<br>(C) Encrypting data (D) Compressing data            | 1 | 1 | 2 |
| 9. Which of the following merges datasets on common index values?<br>(A) merge() (B) merge_on_index()<br>(C) join() (D) concat()                                | 1 | 1 | 2 |
| 10. Which function in Pandas helps in handling missing data?<br>(A) fillna() (B) replace()<br>(C) dropna() (D) fillna() and dropna()                            | 1 | 1 | 3 |
| 11. Which technique is used to group data into bins?<br>(A) Binning (B) Sorting<br>(C) Indexing (D) Concatenation                                               | 1 | 1 | 3 |

|                                                                                                                                                                            |   |   |   |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---|
| 12. What is the process of converting raw data into a clean dataset called?<br>(A) Data Loading (B) Data Transformation<br>(C) Data Filtering (D) Data Encoding            | 1 | 1 | 4 |
| 13. What is the purpose of pivot() in Pandas?<br>(A) To flatten arrays (B) To create a pivot table<br>(C) To merge datasets (D) To clean null values                       | 1 | 1 | 1 |
| 14. Standardization refers to _____.<br>(A) Scaling data to [0, 1] range (B) Scaling data to have mean 0 and variance 1<br>(C) Dropping outliers (D) Merging datasets      | 1 | 1 | 2 |
| 15. Which of the following is a plotting library in Python?<br>(A) Matplotlib (B) NumPy<br>(C) Scikit-learn (D) Pandas                                                     | 1 | 1 | 4 |
| 16. What does a box plot help visualize?<br>(A) Mean and variance only (B) Distribution and outliers<br>(C) Only correlation (D) Pairwise relations                        | 1 | 1 | 4 |
| 17. What is the use of subplots() in Matplotlib?<br>(A) For styling plots (B) For creating multiple plots in one figure<br>(C) For saving plots (D) For creating box plots | 1 | 1 | 4 |
| 18. Which Seaborn plot is best suited for pairwise relationships?<br>(A) scatterplot() (B) pairplot()<br>(C) barplot() (D) boxplot()                                       | 1 | 1 | 4 |
| 19. Which function helps annotate text inside a Matplotlib plot?<br>(A) text() (B) annotate()<br>(C) plot_text() (D) label()                                               | 1 | 1 | 4 |
| 20. Which type of plot is commonly used to visualize frequency distribution?<br>(A) Line plot (B) Histogram<br>(C) Scatter plot (D) Pie chart                              | 1 | 1 | 4 |

**PART - B (4 × 10 = 40 Marks)**

Answer **any 4** Questions

|                                                                                                                      | Marks | BL | CO |
|----------------------------------------------------------------------------------------------------------------------|-------|----|----|
| 21. Discuss the key steps involved in Data Science Process with neat sketch.                                         | 10    | 2  | 1  |
| 22. Analyze the various techniques to handle large datasets and list the challenges faced during data wrangling.     | 10    | 4  | 2  |
| 23. Develop a Python code to clean and transform data using Pandas functions like dropna(), fillna(), and replace(). | 10    | 6  | 3  |
| 24. Discuss briefly about line plots, scatter plots, box plot and histogram used in data visualization.              | 10    | 1  | 4  |
| 25. Categorize the steps involved in merging and concatenating datasets using Pandas.                                | 10    | 4  | 3  |
| 26. Examine how to handle missing data and common techniques to deal with missing values.                            | 10    | 4  | 3  |

**PART - C (1 × 15 = 15 Marks)**

Answer **any 1** Questions

|                                                                                                                                      | Marks | BL | CO |
|--------------------------------------------------------------------------------------------------------------------------------------|-------|----|----|
| 27. Explain various techniques used in data visualization and develop a Python program to create a 3D surface plot using Matplotlib. | 15    | 2  | 4  |
| 28. Discuss in detail about the data cleaning and wrangling techniques on a sample dataset on your choice using Pandas.              | 15    | 2  | 3  |